Analysis and Implementation of g_m/I_D Methodology*

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1 Introduction to the g_m/I_D Methodology

Analog CMOS circuit design traditionally relied on the old-school square-law model for transistor sizing during practical design stages. In modern deep-submicron process nodes, however, short-channel effects (e.g., velocity saturation, mobility degradation, and channel-length modulation) make this model increasingly inaccurate. Furthermore, some modern low-power design requirements frequently force transistors to operate in the moderate and weak inversion regions, where the square-law model fails completely. To overcome these limitations, the g_m/I_D methodology serves as an efficient sizing framework nowadays without losing much physical insight of the transistor itself.

1.1 Transistor Efficiency (TE)

We denote *Transistor Efficiency* (TE) as the ratio of the transconductance g_m to the DC bias current I_D:

$$TE = \frac{g_m}{I_D} \quad \left[\mathrm{V}^{-1}\right] \tag{1}$$

This ratio quantifies how effectively a transistor converts **DC quiescent power** into **small-signal transconductance**. Since we are already familiar with

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \tag{2}$$

it is easy to see that

$$\frac{\partial \ln(I_D)}{\partial V_{GS}} = \frac{1}{I_D} \cdot \frac{\partial I_D}{\partial V_{GS}} = \frac{g_m}{I_D} \tag{3}$$

which offers a direct metric of device operation independent of specific model parameters.

1.2 Performance Trade-Offs: Speed and Noise

The choice of g_m turns out to be the *knob* for both circuit speed and thermal noise performance:

*F. Silveira, D. Flandre, and P. G. A. Jespers, A g_m/I_D Based Methodology for the Design of CMOS Analog Circuits and Its Application to the Synthesis of a Silicon-on-Insulator Micropower OTA, *IEEE Journal of Solid-State Circuits*, vol. 31, no. 9, pp. 1314–1319, Sept. 1996.

- **Speed (Gain-Bandwidth Product):** In a canonical single-pole amplifier loaded by a capacitor C_L, the voltage gain is $A_v = g_m r_o$ and the bandwidth is $f_{3\text{dB}} = 1/(r_o C_L)$. The Gain-Bandwidth Product (GBW) simplifies to:

$$GBW = A_v \cdot f_{3\text{dB}} = g_m r_o \frac{1}{2\pi C_L r_o} = \frac{g_m}{2\pi C_L} \tag{4}$$

  Thus, g_m alone dictates the **high-frequency responsiveness** under negative feedback.

- **Thermal Noise:** The *input-referred* thermal noise spectral density of a MOSFET is inversely proportional to g_m:

$$\overline{v_{in,n}^2} = \frac{\overline{i_n^2}}{g_m^2} = \frac{4kT\gamma g_m}{g_m^2} = \frac{4kT\gamma}{g_m} \tag{5}$$

  Maximizing g_m minimizes thermal noise floor penalties. [1]

1.3 Limitations of Classical BJT and Square-Law Models

In Bipolar Junction Transistors (BJTs), the collector current exhibits an exponential relationship $I_C \propto \exp(V_{BE}/V_T)$, resulting in a **constant** efficiency factor:

$$\left(\frac{g_m}{I_C}\right)_{\text{BJT}} = \frac{1}{V_T} \approx 38.6 \text{ V}^{-1} \quad \text{at } T = 300 \text{ K} \tag{6}$$

Hence, a g_m/I_C-based sizing strategy is unnecessary.

Conversely, for *long-channel* MOSFETs governed by the classical square-law model in strong inversion:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2 \implies \frac{g_m}{I_D} = \frac{2}{V_{ov}} \tag{7}$$

where $V_{ov} = V_{GS} - V_{TH}$ is the overdrive voltage. The classical model predicts that $g_m/I_D \to \infty$ as $V_{ov} \to 0$, which is physically impossible.

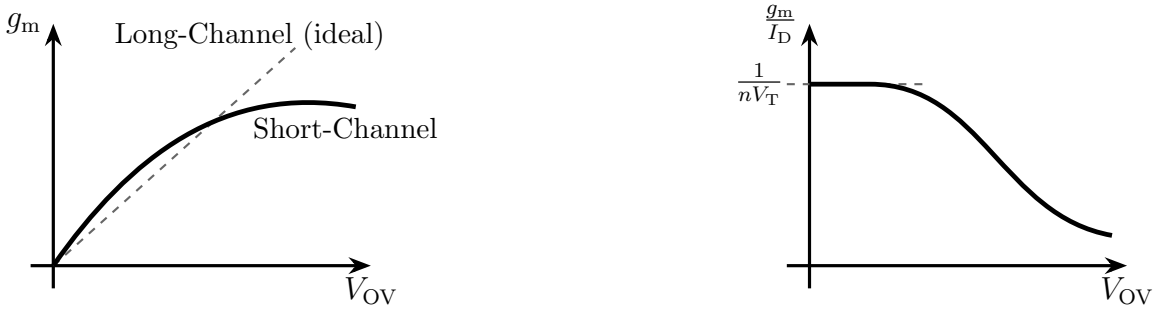


Figure 1: Short-channel MOSFET characteristics showing g_m saturation (left) and g_m/I_D efficiency bound (right).

In actual *short-channel* MOSFETs, g_m saturates at high overdrive voltages due to mobility degradation, while at low overdrive voltages, g_m/I_D flattens out to a maximum upper bound set by subthreshold conduction (weak inversion). Taking the logarithmic derivative of exponential diffusion current yields a hard physical ceiling:

$$\left(\frac{g_m}{I_D}\right)_{\text{max}} = \frac{1}{nV_T}$$

where n is the subthreshold slope factor ($n \approx 1.1 - 1.4$) and $V_T = kT/q \approx 26$ mV. This caps maximum achievable efficiency at around $25 - 35$ V^{-1} in real devices. These effects are depicted in Figure 1.

[1] Note that the *output* thermal noise increases with g_m, following $\overline{i_{out,n}^2} = 4kT\gamma g_m$.

1.4 Regions of Inversion

The value of g_m/I_D directly indicates the operating regime of the MOSFET, shown in Figure 2. Table 1 provides a rule of thumb of choosing the values of g_m/I_D of transistors.

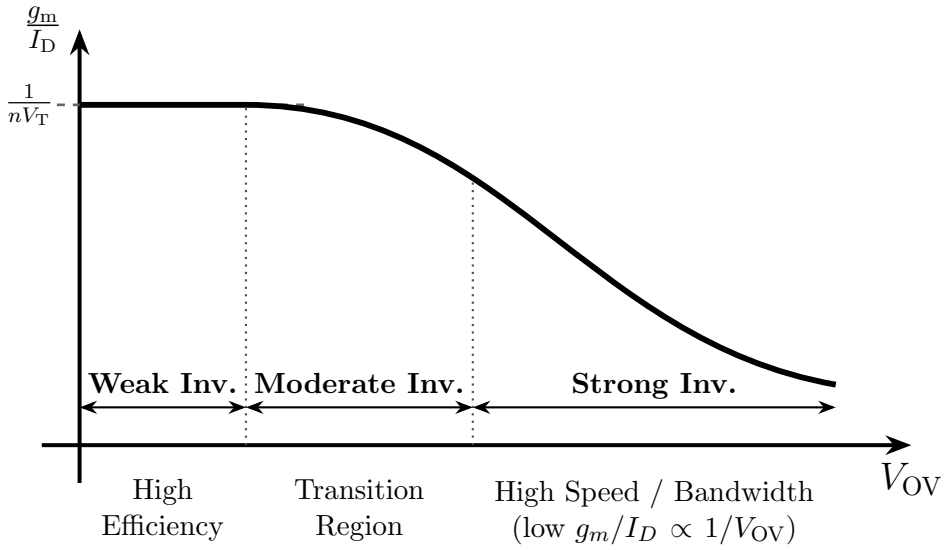


Figure 2: g_m/I_D across weak, moderate, and strong inversion operating regimes.

Table 1: MOSFET Operating Regions Characterized by g_m/I_D

Region	g_m/I_D $(\mathbf{V}^{-1})$	**Trade-Off**	**Applications**
Weak Inversion	> 20	Max g_m/I_D, Min Speed	High-Gain
Moderate Inversion	$10 - 20$	Balanced Efficiency & Speed	General Design
Strong Inversion	< 10	High Speed, Min g_m/I_D	High-Frequency

2 Design Example: Single-Stage OTA

The operational transconductance amplifier (OTA) shown in Figure 3 is specifically designed to drive a load capacitance $C_L = 10$ pF under a strict tail current budget of $I_{\text{tail}} = 1\,\mu$A ($I_{D1,2} = 0.5\,\mu$A per branch). The performance specifications require an open-loop DC gain $A_v > 300$, a phase margin PM $> 60°$, and a Gain-Bandwidth Product target of GBWP > 200 kHz.

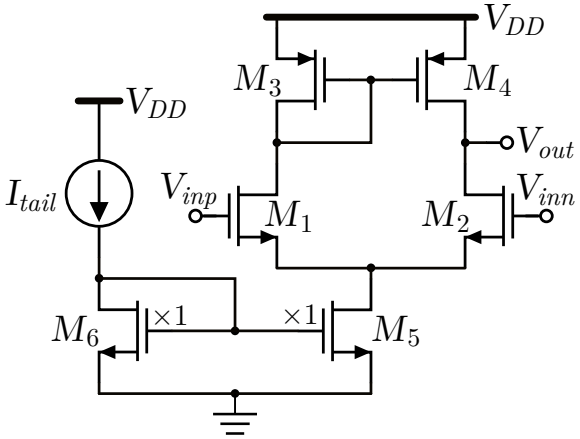


Figure 3: A simple single-stage OTA.

To achieve GBWP > 200 kHz given $C_L = 10$ pF, the required transconductance is:

$$g_{m1} = 2\pi \cdot (200 \text{ kHz}) \cdot (10 \text{ pF}) \approx 12.57\,\mu\text{S} \tag{8}$$

With $I_{D1} = 0.5\,\mu$A, we bias the input pair (M_1, M_2) in deep weak inversion at $(g_m/I_D)_1 = 25$ V^{-1}, pushing transconductance efficiency near its theoretical limit [2].

2.1 Determining Channel Length (L)

To achieve $A_v > 300$, the intrinsic gain requirement remains:

$$A_v = \frac{g_{m1}}{g_{ds1} + g_{ds3}} \approx \frac{1}{2}\left(\frac{g_m}{g_{ds}}\right)_{\text{intrinsic}} \implies \left(\frac{g_m}{g_{ds}}\right)_{\text{intrinsic}} > 600 \tag{9}$$

Evaluating the intrinsic gain curves (Figure 4) across g_m/I_D operating points up to 30 V^{-1}, $L = 2\,\mu$m provides an intrinsic gain of ≈ 1100 at $(g_m/I_D)_1 = 25$ V^{-1}, comfortably satisfying the specification.

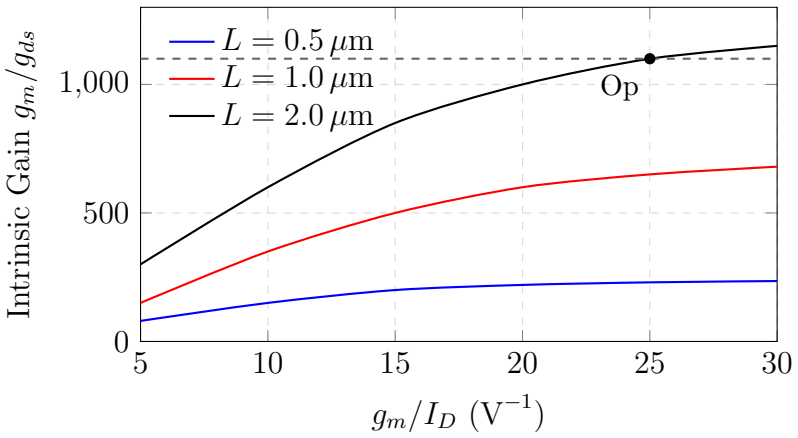


Figure 4: Intrinsic gain g_m/g_{ds} vs. g_m/I_D characterizing channel length L.

2.2 g_m/I_D Operating Point Selection

The g_m/I_D assigned to each transistor pair is reasoned as follows:

- **Input Differential Pair (M_1, M_2) — $(g_m/I_D)_1 = 25$ V^{-1}:** Biasing the input pair at 25 V^{-1} pushes M_1 and M_2 into deep weak inversion, maximizing the transconductance extracted from a limited tail current ($I_{D1} = 0.5\,\mu$A). This is essential to achieve $g_{m1} \approx 12.57\,\mu$S, which directly sets the gain-bandwidth product (GBWP $= g_{m1}/2\pi C_L >$ 200 kHz) without violating the $1\,\mu$A total power constraint.
- **Active Load (M_3, M_4) — $(g_m/I_D)_3 = 10$ V^{-1}:** Selecting a lower efficiency of 10 V^{-1} for the PMOS active load forces M_3 and M_4 deeper into moderate inversion. This reduces both output noise and threshold voltage mismatch.
- **Tail Current Mirror (M_5, M_6) — $(g_m/I_D)_5 = 8$ V^{-1}:** Biasing M_5 and M_6 in strong inversion at 8 V^{-1} increases the required gate overdrive (V_{OV5}). While this consumes a small portion of the output voltage headroom, it substantially boosts the output resistance (r_{ds5}) and reduces V_{th} mismatch, ensuring high common-mode rejection ratio (CMRR) and stable tail current replication.

[2] In modern technology nodes, this limit is generally above 30 V^{-1}.

2.3 Transistor Sizing via Current Density (I_D/W)

With $L = 2\,\mu$m fixed for all devices, we examine the current density curve shown in Figure 5:

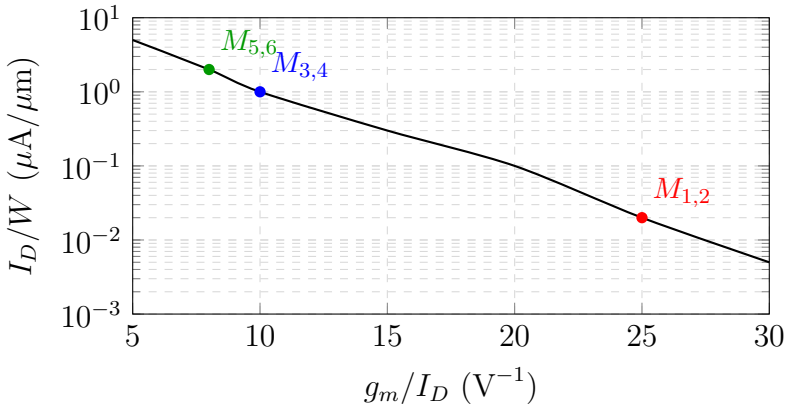


Figure 5: Current density lookup curve (I_D/W vs. g_m/I_D) for $L = 2\,\mu$m.

- M_1, M_2**:** Biased at $(g_m/I_D)_1 = 25$ V^{-1}, corresponding to $I_D/W = 0.02\,\mu$A/μm.
- M_3, M_4**:** Biased at $(g_m/I_D)_3 = 10$ V^{-1}, corresponding to $I_D/W = 1.0\,\mu$A/μm.
- M_5, M_6**:** Biased at $(g_m/I_D)_5 = 8$ V^{-1}, corresponding to $I_D/W = 2.0\,\mu$A/μm.

2.4 Design Table

Final transistor geometry calculations using $W = I_D/(I_D/W)$:

Transistor Pair	g_m/I_D [**V^{-1}**]	I_D/W [μ**A/**μ**m**]	I_D [μ**A**]	**Width** W [μ**m**]
Input Pair (M_1, M_2)	25.0	0.02	0.5	**25.0**
Active Load (M_3, M_4)	10.0	1.0	0.5	**0.5**
Tail Mirror (M_5, M_6)	8.0	2.0	1.0	**0.5**

Table 2: Final OTA Transistor Sizing based on g_m/I_D extraction.

During practical design, readers are encouraged to complete this table from left to right.

2.5 Specification Verification and Design Analysis

To ensure design compliance, the calculated operating points are verified against the target specifications:

- **DC Gain ($A_v > 300$):** With $L = 2\,\mu$m, the intrinsic gain of the output stage yields $(g_m/g_{ds})_{\text{intrinsic}} \approx 1100$. The open-loop gain is $A_v \approx \frac{1}{2}(1100) = 550$, exceeding the > 300 requirement.
- **Gain-Bandwidth Product (GBWP > 200 kHz):** Operating at $(g_m/I_D)_1 = 25$ V^{-1} with $I_{D1} = 0.5\,\mu$A gives $g_{m1} = 12.5\ \mu$S. Driving $C_L = 10$ pF yields GBWP $= \frac{12.5\ \mu\text{S}}{2\pi \cdot 10\ \text{pF}} \approx 198.9$ kHz ≈ 200 kHz, meeting the target at the physical boundary.
- **Phase Margin (PM $> 60°$):** As a single-stage uncompensated OTA driving a dominant capacitive load, the non-dominant high-frequency poles (at the active load mirror node and input tail node) reside well beyond the gain-bandwidth frequency. This guarantees an inherently stable phase margin of PM $\approx 85°$–$90°$.

2.6 Potential Weaknesses and Trade-offs

While the design satisfies all DC and AC metrics within the $1\,\mu$A power budget, two critical non-idealities should be monitored:

1. **Extreme Sensitivity to Subthreshold Mismatch and Flicker Noise:** Biasing M_1, M_2 at $(g_m/I_D)_1 = 25\ \mathrm{V}^{-1}$ pushes the pair into deep weak inversion, where the drain current depends exponentially on threshold voltage variations (V_{th}). Micro-variations in fabrication will manifest as significant input offset voltages (V_{os}). Furthermore, the large channel area ($W/L = 25/2$) increases gate capacitance, slightly adding parasitic loading to the input source.

2. **Limited Slew Rate (SR):** Under a fixed $1\,\mu$A total tail current, the maximum large-signal charging current available to the load capacitor is I_{tail}. The resulting slew rate is:
$$\mathrm{SR} = \frac{I_{\text{tail}}}{C_L} = \frac{1\,\mu\mathrm{A}}{10\ \mathrm{pF}} = 0.1\ \mathrm{V}/\mu\mathrm{s} \tag{10}$$
This severely restricts large-signal transient response speed, making the topology suitable primarily for small-signal processing applications.